

A Countably Generated Counterexample to First Arithmetic-Mean-at-Infinity Rigidity

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Status. Full counterexample, likely valid, pending human review. There is a countably generated operator ideal $I \subset \mathcal{L}_1$ which is not $\text{am-}\infty$ stable although I_{a_∞} is $\text{am-}\infty$ stable. This gives a negative answer to the question following Theorem 5.2 of Kaftal–Weiss.

1 Source question

For a positive decreasing summable sequence $x = (x_n)$ write

$$(Tx)_n = (x_{a_\infty})_n := \frac{1}{n} \sum_{j>n} x_j,$$

and let (x) denote the principal $B(H)$ -ideal generated by x . Kaftal and Weiss define the arithmetic mean at infinity of an ideal I by

$$\Sigma(I_{a_\infty}) = \{z \in c_0^* : z = O(y_{a_\infty}) \text{ for some } y \in \Sigma(I \cap \mathcal{L}_1)\}.$$

For summable x they prove the useful identity

$$(x)_{a_\infty} = (x_{a_\infty}). \tag{1}$$

An ideal J is $\text{am-}\infty$ stable precisely when $J \subset \mathcal{L}_1$ and $J = J_{a_\infty}$ [1, Section 2 and Theorem 5.2]; see also [2, Lemma 4.7 and Corollary 4.10].

The source proves that stability of I_{a_∞} forces stability of I when I is principal, and asks whether this remains true for arbitrary ideals [1, p. 23]:

It may be of interest to note that Theorem 5.1 does not extend beyond principal ideals. In fact, as the two examples below show, it can fail even for countably generated ideals. It is an open question whether or not Theorem 5.2 extends beyond principal ideals, indeed, even whether or not $\text{am-}\infty$ stability of I_{a_∞} implies $\text{am-}\infty$ stability of I .

Example 5.4. *A countably generated ideal N which is not am-stable but for which ${}_a N = N^o = N_- = st^a((\omega))$ is am-stable .*

Figure 1: The open question following Theorem 5.2 in [1, p. 23].

2 Idea of the proof

The tail operator has a convenient slowly varying scale. Put, up to harmless finite initial terms,

$$u_n^{(p)} = \frac{e^{-\sqrt{\log n}}(\log n)^{p/2}}{n} \quad (p = 0, 1, 2, \dots).$$

Integration shows that $Tu^{(p)} \asymp u^{(p+1)}$. Hence the directed union $S = \bigcup_p (u^{(p)})$ is am- ∞ stable.

To obtain a proper ideal with arithmetic mean at infinity equal to S , we flatten every $u^{(p)}$ on a common sparse collection of intervals $[a_k, b_k]$. The intervals are enormous in multiplicative length, $b_k/a_k \asymp (\log a_k)^{2k}$, so at their left endpoints every fixed finite sum of flattened sequences is far too small to dominate $u^{(0)}$, even after any fixed ampliation. At the same time they are tiny on the natural logarithmic tail scale:

$$\frac{\log(b_k/a_k)}{\sqrt{\log a_k}} \longrightarrow 0.$$

Thus flattening removes only a summably small fraction of each tail, and it does not change the principal ideal generated by the tail mean. The ideal generated by the flattened sequences is therefore proper in S , while one application of a_∞ recovers all of S .

3 Construction and estimates

For $p \geq 0$ and $n \geq 3$ set

$$v_n^{(p)} := \frac{e^{-\sqrt{\log n}}(\log n)^{p/2}}{n}, \quad u_n^{(p)} := \sup_{j \geq n} v_j^{(p)}.$$

Then $u^{(p)} \in (\ell^1)^*$, it is decreasing by construction, and it coincides with $v^{(p)}$ from some index onward. Changing the first two terms in any decreasing positive way is understood. Let

$$A_p(n) := \sum_{j > n} u_j^{(p)}.$$

Lemma 1 (tail scale). *For every fixed $p \geq 0$,*

$$A_p(n) \sim 2e^{-\sqrt{\log n}}(\log n)^{(p+1)/2}, \quad Tu^{(p)} \asymp u^{(p+1)}. \quad (2)$$

Moreover, for fixed integers $p, k \geq 0$, if $b = \lceil a(\log a)^{2k} \rceil$, then

$$\frac{A_p(b)}{A_p(a-1)} \longrightarrow 1 \quad (a \rightarrow \infty). \quad (3)$$

Proof. The integral test, followed by $t = \log x$ and $r = \sqrt{t}$, gives

$$\begin{aligned} \sum_{j > n} v_j^{(p)} &\sim \int_n^\infty \frac{e^{-\sqrt{\log x}}(\log x)^{p/2}}{x} dx \\ &= 2 \int_{\sqrt{\log n}}^\infty e^{-r} r^{p+1} dr \sim 2e^{-\sqrt{\log n}}(\log n)^{(p+1)/2}. \end{aligned}$$

The finite monotone-envelope correction does not affect this asymptotic. Dividing by n proves the second assertion in (2). Finally,

$$\sqrt{\log b} - \sqrt{\log a} = O\left(\frac{k \log \log a}{\sqrt{\log a}}\right) = o(1), \quad \frac{\log b}{\log a} \rightarrow 1,$$

so the ratio of the two expressions in (2) tends to one. This proves (3). $\square$

Let $\varepsilon_k = 2^{-k-4}$. By Lemma 1, choose integers a_k inductively and put

$$R_k = (\log a_k)^{2k}, \quad b_k = \lceil a_k R_k \rceil,$$

so that the intervals $[a_k, b_k]$ are disjoint, $\log b_k \leq 2 \log a_k$, all $u^{(p)}$ with $p \leq k$ already equal their displayed formulas after a_k , and

$$A_p(b_k) \geq (1 - \varepsilon_k)A_p(a_k - 1) \quad (0 \leq p \leq k). \quad (4)$$

There are only finitely many requirements at stage k , so the induction is possible.

For every $p \geq 0$, define the decreasing summable sequence $\xi^{(p)}$ by

$$\xi_n^{(p)} := \begin{cases} u_{b_k}^{(p)}, & a_k \leq n \leq b_k \text{ for some } k \geq p, \\ u_n^{(p)}, & \text{otherwise.} \end{cases} \quad (5)$$

Monotonicity is immediate: on a flattened interval the sequence drops to its right-endpoint value and rejoins $u^{(p)}$ there.

Lemma 2 (flattening preserves every tail ideal). *For each $p \geq 0$,*

$$\sum_{j>n} \xi_j^{(p)} \asymp A_p(n), \quad T\xi^{(p)} \asymp u^{(p+1)}, \quad (\xi^{(p)})_{a_\infty} = (u^{(p+1)}).$$

Proof. The mass removed from the k th interval is at most

$$L_{p,k} := \sum_{j=a_k}^{b_k} (u_j^{(p)} - u_{b_k}^{(p)}) \leq A_p(a_k - 1) - A_p(b_k) \leq \varepsilon_k A_p(a_k - 1).$$

If n is outside a flattened interval, every interval contributing to the tail starts after n , and hence the total removed tail is at most

$$A_p(n) \sum_{k \geq 1} \varepsilon_k \leq \frac{1}{16} A_p(n).$$

If $n \in [a_k, b_k]$, discard the nonnegative remainder of the current plateau. The tail after b_k , less all later losses, is at least

$$A_p(b_k) \left(1 - \sum_{q>k} \varepsilon_q \right) \geq (1 - \varepsilon_k) \frac{15}{16} A_p(a_k - 1) \geq (1 - \varepsilon_k) \frac{15}{16} A_p(n).$$

The reverse inequality follows from $\xi^{(p)} \leq u^{(p)}$. Thus the tails are comparable uniformly in n . Lemma 1 now gives $T\xi^{(p)} \asymp u^{(p+1)}$, and (1) gives the principal-ideal identity. $\square$

4 The counterexample

Define the two countably generated ideals

$$I := \sum_{p \geq 0} (\xi^{(p)}) = \bigcup_{P \geq 0} \left(\sum_{p=0}^P \xi^{(p)} \right), \quad S := \sum_{p \geq 0} (u^{(p)}) = \bigcup_{p \geq 0} (u^{(p)}). \quad (6)$$

The sums in (6) are ideal sums; finite sums of principal ideals are principal. Since $\xi^{(p)} \leq u^{(p)}$, one has $I \subset S \subset \mathcal{L}_1$.

Theorem 1. *The ideal I in (6) is not $\text{am-}\infty$ stable, but*

$$I_{a_\infty} = S$$

and S is $\text{am-}\infty$ stable. In particular, $\text{am-}\infty$ stability of I_{a_∞} does not imply $\text{am-}\infty$ stability of I , even for a countably generated ideal $I \subset \mathcal{L}_1$.

Proof. First, $(u^{(p)}) \subset (u^{(p+1)})$ for every p , because $u_n^{(p+1)}/u_n^{(p)} \sim \sqrt{\log n}$. By Lemma 1 and (1),

$$(u^{(p)})_{a_\infty} = (u^{(p+1)}).$$

It follows directly from the definition of I_{a_∞} (or by applying it to the directed union) that

$$S_{a_\infty} = \bigcup_{p \geq 0} (u^{(p)})_{a_\infty} = \bigcup_{p \geq 0} (u^{(p+1)}) = S.$$

Since $S \subset \mathcal{L}_1$, this is $\text{am-}\infty$ stability.

Lemma 2 and monotonicity give, for every p ,

$$(u^{(p+1)}) = (\xi^{(p)})_{a_\infty} \subset I_{a_\infty}.$$

The union of the left sides is S , so $S \subset I_{a_\infty}$. Conversely, $I \subset S$ and monotonicity, followed by stability of S , give $I_{a_\infty} \subset S_{a_\infty} = S$. Therefore $I_{a_\infty} = S$.

It remains to prove that $I \neq S$. We show that $u^{(0)} \notin \Sigma(I)$. Put $g_P = \sum_{p=0}^P \xi^{(p)}$. If $u^{(0)} \in (g_P)$, then for some $m \in \mathbb{N}$ one would have $u^{(0)} = O(D_m g_P)$, where $(D_m x)_n = x_{\lceil n/m \rceil}$. Take $k \geq \max\{P, m\}$ and evaluate at $n = ma_k$. All the first $P+1$ sequences are flat at a_k , so

$$(D_m g_P)_{ma_k} = g_P(a_k) = \sum_{p=0}^P u_{b_k}^{(p)} \leq (P+1)u_{b_k}^{(P)}.$$

For large k , using $b_k/a_k \geq R_k$, $b_k > ma_k$, and $\log b_k \leq 2 \log a_k$, we obtain

$$\begin{aligned} \frac{u_{ma_k}^{(0)}}{(D_m g_P)_{ma_k}} &\geq \frac{R_k}{m(P+1)(\log b_k)^{P/2}} \exp\left(\sqrt{\log b_k} - \sqrt{\log(ma_k)}\right) \\ &\geq \frac{(\log a_k)^{2k-P/2}}{m(P+1)2^{P/2}} \rightarrow \infty. \end{aligned}$$

Thus $u^{(0)} \notin (g_P)$ for every P , and hence $u^{(0)} \notin I$. But $u^{(0)} \in S = I_{a_\infty}$, so $I \neq I_{a_\infty}$. As $I \subset \mathcal{L}_1$, the stability criterion quoted above shows that I is not $\text{am-}\infty$ stable. $\square$

5 Verification, scope, and novelty audit

The construction is exact and has no computational dependency. The main review points are: (1) the standard principal identity (1); (2) the uniform accounting of future tail loss in Lemma 2; and (3) the use of one common family of flattened intervals for all $p \leq k$, which is what makes the final diagonal noncontainment work simultaneously for every finite sum and every fixed ampliation.

The result answers only the specific weakest question displayed in the source: stability of I_{a_∞} need not descend to I . It therefore also shows that the full seven-ideal equivalence of Theorem 5.2 cannot extend verbatim to arbitrary ideals. It does not classify which additional completeness, softness, or lattice hypotheses restore rigidity.

A bounded literature audit searched the exact source question, the phrases “am- ∞ stability” and “arithmetic mean at infinity”, the source authors, arXiv:0707.3115, and the closely related papers arXiv:0707.3169, 0707.3217, 0707.3271, and 1201.4826. No later paper explicitly answering this question was located. The source paper and the run’s lightweight result, attempt, and proof-gap indexes contained no prior solution packet for this target. Novelty is therefore plausible but remains pending specialist review.

References

- [1] V. Kaftal and G. Weiss, *$B(H)$ lattices, density and arithmetic mean ideals*, Houston J. Math. **37** (2011), 233–283; arXiv:0707.3115.
- [2] V. Kaftal and G. Weiss, *Traces on operator ideals and arithmetic means*, J. Operator Theory **63** (2010), 3–46; arXiv:0707.3169.